

Elliptic Laurent periods and A216584

with recipes for reduction certificates

Bradley Klee, Mech.An.ika Sol (Open AI)

August 8, 2026

Abstract

The sequence $a(n) = C_n T_n$ where $C_n = [(x + 2 + x^{-1})^n]_0$ and $T_n = [(y + 1 + y^{-1})^n]_0$ gives the product of central binomial coefficients C_n with central trinomial coefficients T_n . The generating function $A(s) = \sum_{n \geq 0} a_n s^n$ is also the period function of an elliptic surface. The ODE constraining $A(s)$ derives from reduction analysis of the period integrand and as well from the definition of C_n and T_n as constant terms of Laurent polynomials. Two approaches yield distinct certificates.

Theorem 1. *The sequence $a(n)$ and its generating function $A(s)$ satisfy all of the following:*

(1) For $G(x, y) = (x + 2 + x^{-1})(1 + y + y^{-1})$,

$$a_n = [G(x, y)^n]_0, \quad A(s) = \left[\frac{1}{1 - sG(x, y)} \right]_0.$$

(2) Given the elliptic curve $E_s = \{ (u, v) : v^2 = u^3 + (1 - 4s)u^2 + 16s^2u \}$ then

$$A(s) = c_Y^{-1} T_Y(ks) \quad \text{where} \quad T(s) = \oint_Y \frac{du}{v} = \frac{du}{\sqrt{u^3 + (1 - 4s)u^2 + 16s^2u}}$$

and $T(s)$ is both an elliptic integral and the period of E_s .

(3) The terms $a(n)$ satisfy a p -recurrence:

$$n^2 a_n = 2(2n - 1)^2 a_{n-1} + 12(2n - 1)(2n - 3) a_{n-2} \quad (n \geq 2),$$

and the generating function satisfies an ordinary differential equation:

$$L \circ A(s) = s(1 - 12s)(1 + 4s)A'' + (1 - 16s - 144s^2)A' - (2 + 36s)A = 0.$$

(4) Approaches (1) and (2) arrive at the result (3) through distinct certificates

$$L \circ \left(\frac{du}{\sqrt{p}} \right) = d_u \left(\frac{R}{p^{3/2}} \right), \quad L \circ \left(\frac{1}{\rho} \right) = D_x \left(\frac{\Xi_x}{\rho^2} \right) + D_y \left(\frac{\Xi_y}{\rho^2} \right),$$

where $\rho = 1 - sG$, $D_x = x\partial_x$ and $D_y = y\partial_y$, and numerators are easier for machines to read. The exact finite certificate data are recorded in the embedded JSON certificates designed for automatic checking.

The initial values are

1, 2, 18, 140, 1330, 12852, 130284, 1348776, ...

1 A216584 as curve geometry

Curve certificate and recurrence

Set

$$L = s(1 - 12s)(1 + 4s)\partial_s^2 + (1 - 16s - 144s^2)\partial_s - (2 + 36s).$$

Direct differentiation of $P^{-1/2}$ gives the exact identity

$$L\left(\frac{du}{\sqrt{P}}\right) = d_u\left(\frac{R(u, s)}{P^{3/2}}\right), \quad (1)$$

where

$$R(u, s) = 4u^2(96s^3 - 48s^2u - 16s^2 + 18su^2 + 24su + 8s + u^2 + u). \quad (2)$$

After clearing $P^{5/2}$, both sides of (1) have identical polynomial numerator. Hence every closed period of E_s satisfies $L\Pi = 0$.

Substitution of $A(s) = \sum a_n s^n$ into the differential equation in Theorem 1(3) gives the recurrence. With $a_0 = 1$ and $a_1 = 2$, it begins

$$1, 2, 18, 140, 1330, 12852, 130284, 1348776, \dots$$

Product factorization and Bala's diagonal

The central trinomial coefficients satisfy

$$nT_n = (2n - 1)T_{n-1} + 3(n - 1)T_{n-2},$$

while

$$\frac{\binom{2n}{n}}{\binom{2n-2}{n-1}} = \frac{2(2n-1)}{n}.$$

Their product therefore satisfies Theorem 1(3); the initial values identify

$$a_n = \binom{2n}{n} T_n. \quad (3)$$

The factorization G makes this identity literal at the Laurent level:

$$[G^n]_0 = [(x + 2 + x^{-1})^n]_0 [(1 + y + y^{-1})^n]_0 = \binom{2n}{n} T_n.$$

Peter Bala recorded on OEIS the diagonal representation

$$a_n = [X^n Y^n](1 + (X + Y)^2 + (X + Y)^4)^n. \quad (4)$$

To see the same factorization directly, write

$$(1 + z + z^2)^n = \sum_k c_{n,k} z^k.$$

In (4), only $k = n$ can contribute to total bidegree (n, n) . Hence

$$[X^n Y^n](\dots)^n = c_{n,n} [X^n Y^n](X + Y)^{2n} = T_n \binom{2n}{n}.$$

Thus G is a factorized constant-term form of the same known product structure.

The Laurent pencil and the elliptic curve

The level equation $1 - sG = 0$ is

$$s(x+1)^2(y^2+y+1) - xy = 0. \quad (5)$$

Define

$$u = -4sy, \quad v = 4sy \frac{x-1}{x+1}. \quad (6)$$

A direct substitution gives

$$v^2 - u^3 - (1 - 4s)u^2 - 16s^2u = \frac{64s^2y}{(x+1)^2} (s(x+1)^2(y^2+y+1) - xy).$$

Thus (6) maps the Laurent pencil birationally to E_s ; away from the exceptional divisors the inverse is

$$y = -\frac{u}{4s}, \quad x = \frac{u-v}{u+v}. \quad (7)$$

The period differential is preserved as well. If $Q = s(x+1)^2(y^2+y+1) - xy$, then

$$\frac{dx/x \wedge dy/y}{1 - sG} = -\frac{dx \wedge dy}{Q}.$$

On $Q = 0$,

$$Q_x = y \frac{x-1}{x+1},$$

and therefore

$$\text{Res}_{Q=0} \frac{dx/x \wedge dy/y}{1 - sG} = -\frac{dy}{Q_x} = \frac{du}{v}. \quad (8)$$

This proves the period identity in Theorem 1(2) directly, not merely by comparison of differential equations.

For orientation, the discriminant of E_s is

$$-4096s^4(1+4s)(12s-1).$$

The minimal rational elliptic surface has fibres I_4, I_1, I_1, I_0^* at $0, -1/4, 1/12, \infty$, respectively. The visible 2-torsion point $(0,0)$ gives the 2-isogenous companion

$$v^2 = u^3 + (8s-2)u^2 + (1-12s)(1+4s)u,$$

whose exact differential-equation certificate is included in the accompanying verification file; it has the same differential operator as Theorem 1(3).

The same second-order equation also controls the other local periods of this surface. In particular, with

$$H(z) = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right),$$

one may take local solutions

$$\Pi_{1/12}(q) = \sqrt{1 + q/3} H(q), \quad q = \frac{1 - 12s}{1 + 4s},$$

and

$$\Pi_{-1/4}(r) = \sqrt{1 - 4r} H(r), \quad r = \frac{1 + 4s}{16s}.$$

The A216584 branch is the analytic solution normalized at $s = 0$; the second local solution there is logarithmic.

2 The Laurent–ODE first approach

We now forget the curve, the termwise product factorization, and the known recurrence, and start only from G . Write

$$\rho = 1 - sG(x, y), \quad \Phi(s) = \left[\frac{1}{\rho} \right]_0.$$

Equivalently,

$$\Phi(s) = \frac{1}{(2\pi i)^2} \oint \oint \frac{dx}{x} \frac{dy}{y} \frac{1}{\rho}. \quad (9)$$

Let $\theta = s\partial_s$. Successive differentiated integrands are

$$\frac{1}{\rho}, \quad \theta \frac{1}{\rho} = \frac{sG}{\rho^2}, \quad \theta^2 \frac{1}{\rho} = \frac{sG}{\rho^2} + \frac{2s^2 G^2}{\rho^3}.$$

The reduction asks whether these classes become linearly dependent modulo D_x - and D_y -divergences.

For this input one may take

$$B = \{x^i y^j : -2 \leq i, j \leq 2\}, \quad C = \{x^k y^\ell : -3 \leq k, \ell \leq 3\}.$$

Thus $|B| = 25$ and $|C| = 49$. Here the basis-enlargement parameter is $d = 2$: explicitly,

$$B = (2 \operatorname{conv}(\{0\} \cup \operatorname{supp} G)) \cap \mathbb{Z}^2 = [-2, 2]^2 \cap \mathbb{Z}^2.$$

Since $\operatorname{supp} G = [-1, 1]^2 \cap \mathbb{Z}^2$, multiplication by $\rho = 1 - sG$ enlarges the exponent region by one unit in each direction, so

$$C = [-3, 3]^2 \cap \mathbb{Z}^2.$$

This d is a support bound and is independent of the pole-order parameter below. No support outside C can occur in the linear system for this chosen ansatz.

The matrices are large only in dimension; their entries are given by a small fixed pattern of exponent shifts. Write

$$f_{r,q} = [x^r y^q]G,$$

with $f_{r,q} = 0$ outside $\{-1, 0, 1\}^2$. For $\alpha = (k, \ell) \in C$ and $\beta = (i, j) \in B$, the three 25-column blocks of G are

$$\begin{aligned} G_{\alpha,\beta}^{(0)} &= \delta_{\alpha,\beta} - s f_{k-i, \ell-j}, \\ G_{\alpha,\beta}^{(x)} &= s(k-i) f_{k-i, \ell-j}, \\ G_{\alpha,\beta}^{(y)} &= s(\ell-j) f_{k-i, \ell-j}. \end{aligned} \tag{10}$$

Hence every column of the first block has at most nine nonzero entries and every column of either derivative block has at most six. In all,

$$G : 49 \times 75, \quad \# \text{nz}(G) = 525,$$

so only 14.3% of its entries are nonzero. The divergence matrix is even simpler: for $\beta = (i, j)$,

$$J_{\beta,\gamma}^{(x)} = i \delta_{\beta,\gamma}, \quad J_{\beta,\gamma}^{(y)} = j \delta_{\beta,\gamma}. \tag{11}$$

Thus $J : 25 \times 50$ has only 40 nonzero entries. Because a second s -derivative produces a numerator over ρ^3 , only one pole-lowering step is needed, with pole-order parameter $r = 2$. The complete fixed order-two system is

$$\begin{pmatrix} G & 0 & 0 \\ I_{25} & -J/2 & -[\rho]_B & -[G]_B \end{pmatrix} \begin{pmatrix} u \\ v_x \\ v_y \\ \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} E[G^2]_B \\ 0 \end{pmatrix}.$$

It has size 74×77 and only 608 nonzero entries (10.7%).

For this example the rational reconstruction is also finitely bounded. The chosen exact solution has numerator degree at most 3 and denominator degree at most 4, hence total degree at most 7. Total-degree caps 5 and 6 fail, whereas 7 succeeds and passes complete symbolic replay. Accordingly the fixed reconstruction used below takes eight regular integer values to determine the rational functions and three further values, not used in that determination, as independent checks. After interpolation every entry is substituted back into the original 74×77 system over $\mathbb{Q}(s)$; the certificate does not rest on the sampled equations.

The resulting differential relation is

$$A_\theta = \theta^2 - 2s(2\theta + 1)^2 - 12s^2(2\theta + 1)(2\theta + 3). \tag{12}$$

In ordinary derivatives, (12) is exactly Theorem 1(3).

The same solve returns the certificate rather than requiring a second calculation. There are Laurent polynomials Ξ_x, Ξ_y , supported on the 25 exponent pairs of B , such that

$$A\left(\frac{1}{\rho}\right) = D_x\left(\frac{\Xi_x}{\rho^2}\right) + D_y\left(\frac{\Xi_y}{\rho^2}\right), \tag{13}$$

where

$$A = (-48s^3 - 8s^2 + s)\partial_s^2 + (-144s^2 - 16s + 1)\partial_s - (36s + 2).$$

This is the same operator as Theorem 1(3). The exact sparse coefficients of Ξ_x, Ξ_y , together with the ordered bases and matrices, are stored in `examples/public/low_support/certificates/A216584_G_guvj.json`. The publication replay checks, by coefficient comparison,

$$\begin{aligned} GUVJ &= 0, \quad X\rho = 0, \\ A(1/\rho) - D_x(\Xi_x/\rho^2) - D_y(\Xi_y/\rho^2) &= 0. \end{aligned}$$

It also regenerates the coefficients and verifies Theorem 1(3). All five exact checks pass. No finite-term fit is used to obtain (12).

Thus the curve calculation of Section 1 and the Laurent reduction of this section meet at the same certified differential equation by independent routes.

3 Formal reductions

The preceding calculation generalizes. Let $F \in K[x^{\pm 1}, y^{\pm 1}]$, write

$$\rho = 1 - tF, \quad D_x = x \frac{\partial}{\partial x}, \quad D_y = y \frac{\partial}{\partial y},$$

and consider

$$\Phi_F(t) = \begin{bmatrix} 1 \\ \rho \end{bmatrix}_0.$$

The constant term of either $D_x(R)$ or $D_y(R)$ is zero. Hence an identity

$$A\left(\frac{1}{\rho}\right) = D_x(\Xi_x) + D_y(\Xi_y) \tag{14}$$

proves $A\Phi_F = 0$.

Fix a pole order r . If Laurent polynomials w, a, b_x, b_y satisfy

$$w = \rho a - D_x(\rho)b_x - D_y(\rho)b_y, \tag{15}$$

then

$$\frac{w}{\rho^{r+1}} = \frac{a - r^{-1}(D_x b_x + D_y b_y)}{\rho^r} + D_x\left(\frac{b_x}{r\rho^r}\right) + D_y\left(\frac{b_y}{r\rho^r}\right). \tag{16}$$

This is the elementary pole-lowering identity behind the calculation.

Let $P = \text{conv}(\{0\} \cup \text{supp } F)$. Choose an integer basis-enlargement parameter $d \geq 1$ and take

$$B_d = (dP) \cap \mathbb{Z}^2.$$

Let C_d be the union of B_d with all translates $B_d + u$, $u \in \text{supp } F$. These are explicit finite exponent bounds: if a, b_x, b_y are supported in B_d , every coefficient equation in (15) lies in C_d . The parameter d controls only this support ansatz; it is distinct from the pole order r in (16). A successful exact replay proves that the selected bound is sufficient for that certificate. We make no claim here that a fixed d works for every Laurent polynomial.

Let $B = (b_1, \dots, b_N)$ be the ordered monomial basis associated with B_d , and let C be the ordered monomial basis associated with C_d . The coefficient matrix

$$G \in K(t)^{M \times 3N}$$

represents the map

$$(a, b_x, b_y) \longmapsto \rho a - D_x(\rho)b_x - D_y(\rho)b_y.$$

If E includes the B -space into the C -space, exact solution of

$$G \begin{pmatrix} u \\ v_x \\ v_y \end{pmatrix} = E \tag{17}$$

produces both the reduced part and the divergence data. The matrix

$$J : (b_x, b_y) \mapsto D_x(b_x) + D_y(b_y)$$

then gives the one-step reduction

$$\text{Lower}_r = U - \frac{1}{r} J \begin{pmatrix} V_x \\ V_y \end{pmatrix}. \quad (18)$$

Linear dependence among finitely many reduced derivative columns gives the differential operator A ; the same linear combination of the stored V -data gives the exact certificate Ξ . The basis-enlargement parameter is independent of the pole order r ; the implementation records both explicitly.

For comparison, the bounded low-support search that led back to A216584 also produced the following small certified OEIS hits:

OEIS	Laurent polynomial	order	exact certificate
A002894	$x^2 + xy + (xy)^{-1} + x^{-2}$	2	yes
A006480	$xy + x/y + x^{-2}$	2	yes
A216584	$(x + 2 + x^{-1})(1 + y + y^{-1})$	2	yes

The first two are retained as regression cases rather than developed here. The same bounded search also produces additional period fingerprints with no OEIS identification in the initial lookup; those are candidates for later geometric study, not claims of new sequences in this paper.

4 A bounded replay for A216584

The general theory above admits many implementation choices. For the main example we remove those choices. The following pseudocode is a specification of one finite calculation, not a heuristic search procedure. It fixes the support enlargement, differential order, pole order, interpolation bound, and validation count in advance. An independent implementation therefore has no operator-order search or basis search to reproduce.

1. Set

$$F = (x + 2 + x^{-1})(1 + y + y^{-1}), \quad \rho = 1 - sF,$$

and use exact rational arithmetic throughout.

2. Fix the support enlargement $d = 2$. Order lexicographically

$$B = [-2, 2]^2 \cap \mathbb{Z}^2, \quad C = [-3, 3]^2 \cap \mathbb{Z}^2.$$

Thus $|B| = 25$, $|C| = 49$. Store a Laurent polynomial as its coefficient vector on one of these ordered sets.

3. Build $G \in \mathbb{Q}(s)^{49 \times 75}$ from

$$(a, b_x, b_y) \mapsto \rho a - D_x(\rho)b_x - D_y(\rho)b_y$$

using the entry formulas (10). Build the inclusion matrix $E \in \mathbb{Q}^{49 \times 25}$ and $J \in \mathbb{Q}^{25 \times 50}$ from (11). Check

$$\#nz(G) = 525, \quad \#nz(J) = 40.$$

4. Form the B -coordinate columns $f = [F]_B$, $f_2 = [F^2]_B$, and $r_0 = [\rho]_B$. Fix differential order two and pole order two. Assemble

$$M = \begin{pmatrix} G & 0 & 0 & -f \\ I_{25} & -J/2 & -r_0 & -f \end{pmatrix}, \quad b = \begin{pmatrix} Ef_2 \\ 0 \end{pmatrix}.$$

Check M has shape 74×77 and 608 nonzero entries. The unknown is $z = (U, V_x, V_y, \alpha, \beta)^T$.

5. At the first regular positive integer value of s , determine pivot columns and an equal number of independent rows. Keep that square pivot block fixed at every later sample. Set all free variables to zero.
6. Fix the rational reconstruction bound

$$\deg(\text{numerator}) + \deg(\text{denominator}) \leq 7.$$

Solve the fixed square block exactly at regular integer values of s . Use eight values to determine the rational functions and three additional regular values only as checks. For each pivot variable reconstruct the least-total-degree rational function satisfying the first eight values and reject it unless it also satisfies all three check values.

7. Reassemble the symbolic vector $z \in \mathbb{Q}(s)^{77}$ and require the full exact identity

$$Mz = b.$$

This symbolic replay, not interpolation, is the proof of the reconstructed solution. Split z into $U, V_x, V_y, \alpha, \beta$.

8. Verify separately

$$G \begin{pmatrix} U \\ V_x \\ V_y \end{pmatrix} = E f_2, \quad U - \frac{1}{2} J \begin{pmatrix} V_x \\ V_y \end{pmatrix} = \alpha r_0 + \beta f.$$

These are the G, U, V, J reduction and the single pole-lowering closure.

9. Form

$$X = \begin{pmatrix} 1 & 0 & 2\alpha \\ 0 & 1 & 2\beta \end{pmatrix}.$$

Take the primitive polynomial kernel vector $p = (p_0, p_1, p_2)^T$, normalized by denominator clearing, polynomial-gcd removal, integer-content removal, and sign fixing. Set

$$A = p_0 + p_1 \partial_s + p_2 \partial_s^2.$$

10. Set $V = (V_x, V_y)^T$ and $\Xi = p_2 V$. Verify the cleared coefficient identity equivalent to

$$A(1/\rho) = D_x(\Xi_x/\rho^2) + D_y(\Xi_y/\rho^2).$$

Require every coefficient in the ordered Laurent basis to vanish exactly.

11. Convert A to $\theta = s \partial_s$ form and obtain the recurrence by replacing each $s^k Q_k(\theta)$ with $Q_k(n-k) a_{n-k}$. The result must be Theorem 1(3).
12. Generate twelve constant terms by exact lattice convolution, replay the recurrence on every available index, and independently compute the first five constant terms by direct expansion. Require both lists to agree.
13. Serialize $B, C, G, J, U, V_x, V_y, \alpha, \beta, A, \Xi$, the recurrence, and the term checks. A verifier must rebuild the coefficient identities from these data rather than trust stored pass flags.

For the exact solution produced by this replay, the largest numerator degree is 3 and the largest denominator degree is 4. The degree-7 reconstruction bound is therefore attained; repeating the same reconstruction with caps 5 or 6 fails, while cap 7 succeeds and the resulting functions pass the full symbolic system. These numerical degree bounds concern reconstruction in s ; they are separate from both the exponent bound $d = 2$ and the pole order $r = 2$.

The accompanying implementation exposes the same calculation through

```
python3 code/guvj/guvj_period_factory.py \
--F "(x + 2 + 1/x)*(1 + y + 1/y)" \
--model A216584_G \
--dilation 2 \
--max-total-degree 7 \
--output examples/public/low_support/certificates/A216584_G_guvj.json
```

```
python3 code/public/verify_a216584_geometry.py
```

The first command builds the Laurent certificate. The second independently checks the elliptic differential identity, recurrence, termwise product, Bala diagonal formula, birational map, residue differential, and agreement with the stored constant terms. The paper deliberately specifies only the A216584 instance in algorithmic detail; Section 3 gives the general reduction from which it is obtained.

Computational disclosure

The symbolic calculations and certificate replays in this paper were performed with the accompanying Python/SymPy implementation. OpenAI language models were used during development for mathematical exploration, code assistance, testing, and editorial work. Claims in the paper do not rely on uncheckable model output: the asserted differential identities, recurrences, birational maps, and Laurent reductions are represented by exact symbolic identities or machine-readable certificates that can be replayed independently.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A216584. The entry includes the product formula of Paul D. Hanna, Peter Bala’s diagonal formula (2020), and Mark van Hoeij’s hypergeometric generating function (2025).
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A002894.
- [3] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A006480.
- [4] B. J. Klee, *An Update on the Computational Theory of Hamiltonian Period Functions*, Ph.D. dissertation, University of Arkansas, Fayetteville, 2020, especially Chapter 3.
- [5] A. Bostan, L. Dumont, and B. Salvy, “Algebraic Diagonals and Walks: Algorithms, Bounds, Complexity,” *Journal of Symbolic Computation* 83 (2017), 68–92.
- [6] A. Bostan, S. Chen, F. Chyzak, and Z. Li, “Complexity of creative telescoping for bivariate rational functions,” *Proceedings of ISSAC 2010*, 203–210.
- [7] G. Almkvist and D. Zeilberger, “The method of differentiating under the integral sign,” *Journal of Symbolic Computation* 10 (1990), 571–591.
- [8] A. Meurer et al., “SymPy: symbolic computing in Python,” *PeerJ Computer Science* 3 (2017), e103.